# CLASS 7 — MATHS · FORMULA SHEET

Chapter 8: **Working with Fractions** | Quick revision, one page

## Multiplying fractions

- $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ (cancel common factors first to save work).
- 'of' means multiply: $\frac{1}{2}$ of $\frac{3}{4} = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$.
- Mixed numbers: convert to improper fractions before multiplying: $1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$.
- Multiplying by a proper fraction makes a number smaller; by an improper fraction (greater than 1) makes it larger.

## Reciprocal

- Reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$ and $\frac{a}{b} \times \frac{b}{a} = 1$.
- Reciprocal of a whole number $n$ is $\frac{1}{n}$. Reciprocal of 1 is 1. Zero has no reciprocal.

## Dividing fractions

- $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ (multiply by the reciprocal of the divisor).
- Whole number ÷ fraction: $4 \div \frac{1}{2} = 4 \times 2 = 8$. Fraction ÷ whole number: $\frac{3}{4} \div 3 = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$.
- Number of pieces = total length ÷ length of one piece.

## Area and word problems

- Area of a rectangle $= l \times b$ with fraction sides, e.g. $\frac{3}{4} \times \frac{2}{5} = \frac{3}{10}$ m$^2$.
- Remaining part $= 1 -$ (fraction used); leftover = total − (pieces × size).

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